

M Tech (AIML)_Machine Learning _Assignment 1

Problem Statement 3

You are part of a research team working on **intelligent environmental sensing systems** deployed across diverse geographies. These systems rely on **low-cost sensors**, which often generate **noisy, inconsistent, and outlier-prone data**.

Your task is to design a **robust multi-class classification pipeline** that can infer the **environmental condition category** from such imperfect sensor readings.

The dataset provided simulates real-world imperfections such as:

- Physically inconsistent measurements
- Extreme outliers
- Mixed data types (continuous + categorical)
- Non-linear feature interactions

Your goal is not just prediction, but to **design a principled ML pipeline that is reliable under data uncertainty**.

1. Data Understanding & Sanity Diagnostics

a. Load the dataset and display a structured summary:

- Schema, data types, and unique value counts
- Sample records (minimum 5)

b. Identify:

- Physically impossible values (e.g., humidity > 100%)
- Suspicious ranges (e.g., extremely high wind speeds)

c. Formulate **at least 3 hypotheses** about potential data quality issues and their implications for model learning.

2. Exploratory Data Analysis with Research Depth

a. Perform **univariate + bivariate + multivariate analysis**:

- Distribution plots (identify skewness/heavy tails)
- Class imbalance visualization
- Interaction plots (pairwise relationships)

b. Conduct **correlation + dependency analysis**:

- Pearson / Spearman correlation
- Mutual Information (mandatory)

c. Answer analytically:

Does linear correlation sufficiently capture feature relationships in this dataset?
Justify using evidence.

d. Identify **at least two feature interactions** that may benefit non-linear models.

3. Robust Data Preprocessing Pipeline

You must design a **defensible preprocessing pipeline**.

a. Handle:

- Missing values (if any)
- Outliers (MANDATORY: compare at least 2 techniques)
 - e.g., IQR capping vs Isolation Forest vs Winsorization

b. Encode categorical variables:

- Compare at least two encoding techniques (e.g., One-Hot vs Target Encoding vs Ordinal)

c. Feature Scaling:

- Compare Standardization vs Robust Scaling vs MinMax
- Justify based on distribution properties

d. Feature Engineering (Compulsory):

- Create at least **2 derived features**, e.g.:
 - Thermal discomfort index
 - Visibility-pressure interaction
 - Seasonal encoding

e. Feature Selection:

- Use **at least two methods**:
 - Filter (Mutual Info / Chi-square)
 - Embedded (Tree-based importance)

4. Model Development & Experimental Design

a. Train-test splits:

- 80:20 (mandatory)
- One additional custom split (justify choice)

b. Implement **at least 4 models**:

- k-NN
- SVM
- Decision Tree

- Naïve Bayes

c. Additionally (IIT-level requirement):

- Include **at least one ensemble model**:
 - Random Forest / Gradient Boosting

d. Hyperparameter tuning:

- Use GridSearchCV or RandomizedSearchCV
- Clearly justify:
 - Parameter ranges
 - Evaluation strategy

e. Cross-validation:

- Use k-fold CV ($k \geq 5$)
- Explain variance in performance

5. Model Evaluation & Comparative Analysis

a. Evaluate using:

- Accuracy
- Precision, Recall, F1-score
- Confusion Matrix
- Macro vs Weighted metrics (important for imbalance)

b. Provide:

- Comparative table of all models
- Ranking based on multiple criteria

c. Analytical questions:

1. Which model is most robust to noise and outliers? Why?
2. Which model benefits most from feature engineering?
3. Does increased complexity always improve performance?

Submission Requirements

- Jupyter notebook(.ipynb) file only
- Clear modular code structure
- Inline comments explaining reasoning
- Each model tagged with contributor name
- Final section: **Contribution Report**